

SNIFFING ATTACK

INTRODUCTION

In the realm of cybersecurity, data confidentiality is a critical concern. While encryption and secure protocols aim to protect data in transit, attackers often exploit network weaknesses using a technique called Sniffing. This report discusses what sniffing is, the various types of sniffing attacks, tools used to perform them, and introduces Wireshark, one of the most powerful sniffing and network analysis tools.

1. What is Sniffing?

Sniffing refers to the process of monitoring and capturing data packets that flow through a network. It's commonly used by network administrators for legitimate purposes such as debugging and traffic analysis. However, in the hands of malicious attackers, sniffing becomes a serious security threat.

In simple terms, a sniffing attack is when a hacker captures unencrypted data being transmitted over the network without the knowledge or permission of the users involved.

2. Types of Sniffing Attacks

Sniffing attacks are generally categorized into two types:

1. Passive Sniffing:

- Monitors traffic without altering or interfering with it.
- Effective only in networks using hubs (where traffic is broadcasted to all).
- Hard to detect since it doesn't generate noticeable activity.
- Targets sensitive information like usernames, passwords, or session IDs.

2. Active Sniffing:

- Actively sends packets or manipulates network behavior to capture more traffic.

- Typically used in switched networks (modern networks using switches instead of hubs).
- Examples include:
 - o **MAC Flooding:** Overloads the switch's MAC table to force it to broadcast all traffic.
 - o **ARP Spoofing/Poisoning:** Fools devices into sending data to the attacker.
 - o **DNS Spoofing:** Redirects DNS queries to attacker-controlled servers.

3, Tools Used for Sniffing Attacks

Below are some popular tools used by attackers and researchers for sniffing:

Tool	Description
Wireshark	Most widely used packet analyzer; supports deep inspection of protocols.
Tcpdump	Command-line packet analyzer for Unix/Linux systems.
Ettrecap	Supports sniffing in switched LANs using ARP poisoning.
dsniff	Collection of tools for password sniffing on networks.
Cain & Abel	Can perform ARP poisoning and sniff VoIP calls, passwords, etc.
MITMf	Man-In-The-Middle attack framework used for sniffing and injecting traffic.
Ngrep	Network version of grep, used to search network traffic for strings.

4. What is Wireshark? Features and Uses

Wireshark is a free and open-source network protocol analyzer that lets you see what's happening on your network at a microscopic level. It is used by security professionals, system administrators, network engineers, and ethical hackers.

Features:

- Captures packets in real time and displays them in human-readable form.

- Supports hundreds of network protocols (TCP, UDP, HTTP, FTP, DNS, etc.).
- Live traffic monitoring and analysis.
- Powerful filtering capabilities.
- Decryption support for many protocols (SSL/TLS).
- Graphical visualizations of network data.

Uses:

- Troubleshooting network issues.
- Detecting malicious traffic and intrusions.
- Analyzing protocol behavior.
- Teaching and learning about network protocols.
- Conducting digital forensics and incident response.

CONCLUSION

Sniffing attacks pose a serious risk to data security, especially in unencrypted or poorly secured networks. Understanding how these attacks work and the tools used by attackers is critical for both defense and prevention. Tools like Wireshark help not only in detecting these attacks but also in understanding how data flows across networks. Organizations must implement strong encryption protocols, monitor traffic actively, and educate users to reduce the risk of sniffing-based intrusions.